Applying Design Patterns in a Microservices-Based Dig

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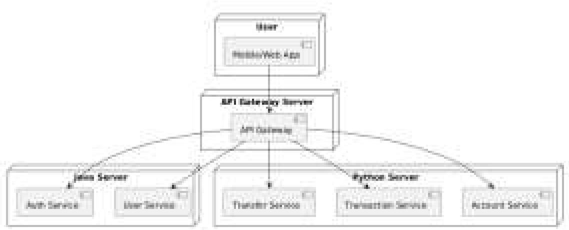
Motivation

Digital financial solutions have emerged to address inefficiencies in traditional banking, such as high fees, slow transactions, and accessibility limitations. This project explores the application of design patterns to enhance scalability and maintainability in a microservices-based digital wallet.

System Architecture

The system consists of two backends: a user authentication service (Java, Spring Boot) and a transaction service (Python, FastAPI). An API Gateway facilitates communication.

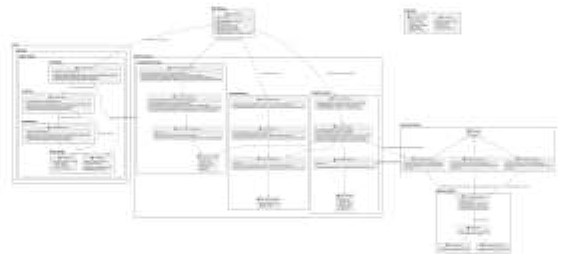
Deployment Diagram



Implementation Details

The microservices interact through a RESTful API. Data is stored using JSON files, and Docker is used for containerization and deployment. Dependencies include FastAPI, Uvicorn, and Python-dotenv.

Class Diagram of the Digital Wallet System



Conclusion

By leveraging design patterns, this digital wallet achieves a structured and scalable architecture. Future enhancements will focus on optimizing transaction handling and incorporating AI-based fraud detection.